CSCI316 - Project 2

Gulf Arabic – English
Code Switched Sentiment Analysis

Transfer Learning for Low-Resource Language Understanding

Group Members:

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Introduction

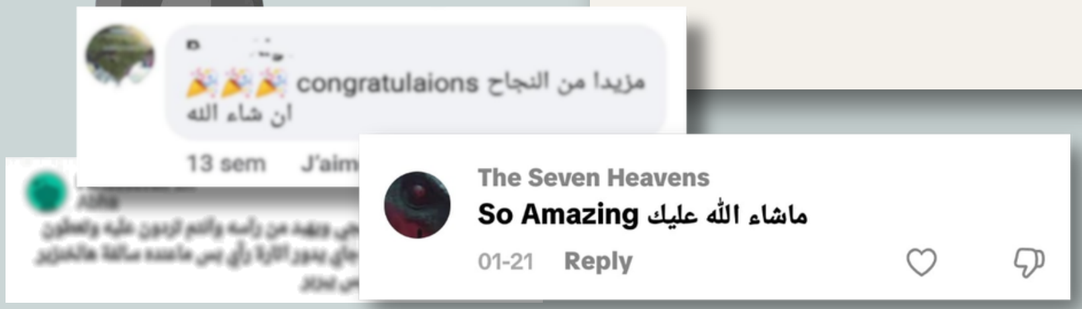
- Natural Language Processing (NLP) has advanced significantly with transformer-based models like BERT and GPT
- These models improve tasks like text classification, translation, and sentiment analysis

- However, most models are heavily trained on English data
- Arabic dialects, especially Gulf Arabic, are underrepresented

In the UAE, social media commonly includes code-switching between Arabic and English
Example: "وما كان زين late الطلب وصل وايد"

Problem Statement

- Traditional sentiment analysis models struggle with:
- Informal Gulf dialects
- Mixed Arabic-English sentences
- Monolingual systems fail to capture full context
- Arabic-only models struggle with English words and vice versa
- This leads to inaccurate sentiment detection in real-world UAE data
- There is a need for a system that understands dialect + code-switching together



Task and Language Justification

Why Gulf Arabic?

- Widely used in everyday communication in the UAE and Gulf region
- Underrepresented in NLP research despite high usage

Why Sentiment Analysis?

- Helps businesses understand customer opinions
- Useful for government policy feedback and social research
- Three classes: neutral is common in Gulf social media – binary would misclassify it

Why Transfer Learning?

- Gulf Arabic datasets are limited
- Pretrained models allow reuse of existing linguistic knowledge
- Reduces data and computational requirements

Dataset Composition

Source

- No single dataset exists for Gulf Arabic–English sentiment analysis
- Dataset created by combining multiple sources
- Total samples: 301,588

Includes:

- Arabic text
- Code-switched text
- Synthetic data to fill gaps

Composition

Data split:

- Training: ~70%
- Validation: ~13%
- Test: ~15%

Code-switched data dominates (76.5%)
Pure Arabic data: 23.5%

Class distribution is imbalanced:

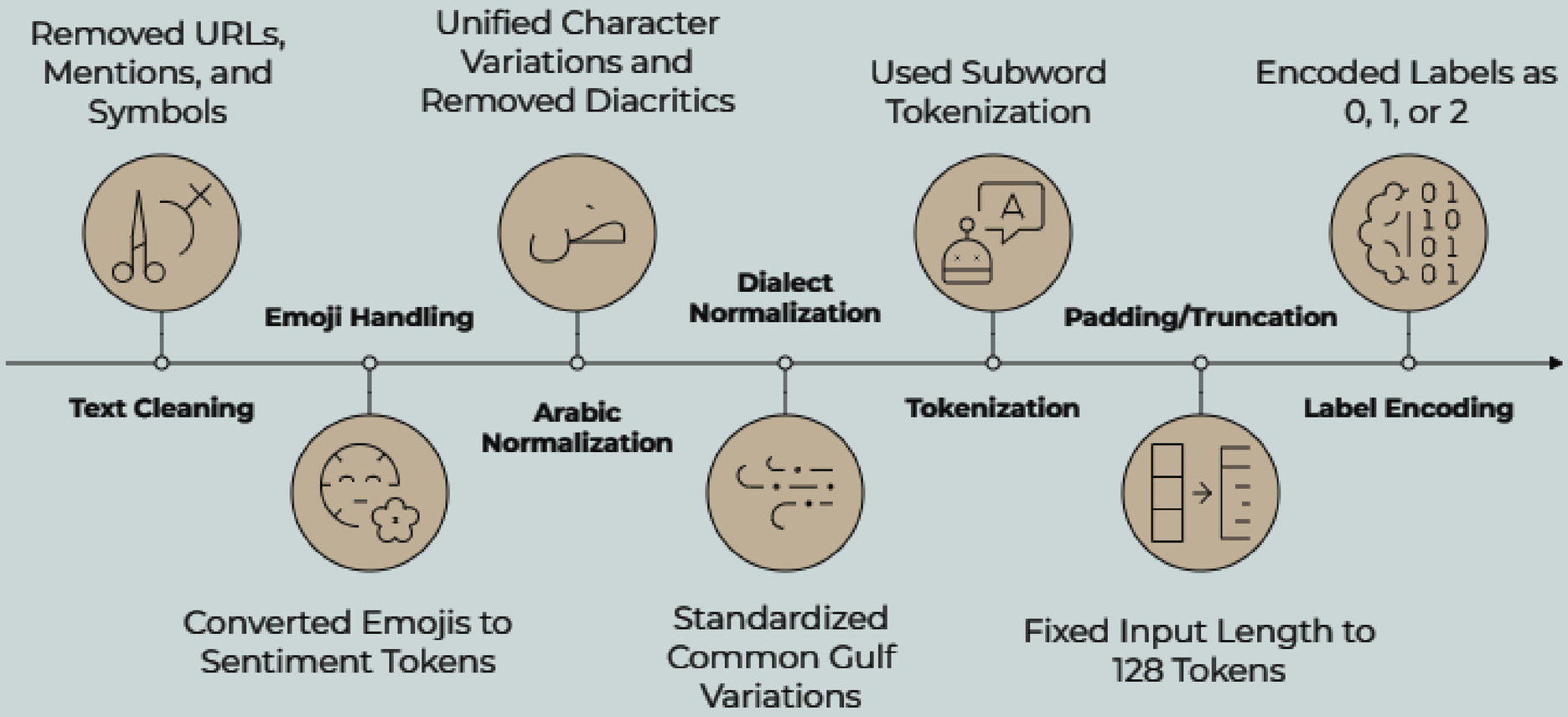
- Positive (38.5%)
- Neutral (36.2%)
- Negative (25.3%)

Macro F1 Score used for fair evaluation

Challenges

- No standardized dataset
- High code-switching (76.5%)
- Complex tokenization
- Synthetic data may not reflect real-world dialect, slang, and context accurately
- Class imbalance - underrepresentation of negative class (25.3%) biases predictions
- Dialect variation in Gulf Arabic
- 64.9% of labels automatically assigned via AraSenti lexicon (weak supervision)
- Twitter API revoked during project - forced reliance on synthetic data

Preprocessing Pipeline



Models & Architecture

LLM selected:

MARBERTv2

Pretrained on 1 billion Arabic tweets | Covers Gulf, Saudi, Egyptian dialects
Arabic-focused model | 91,359 Arabic token IDs in vocabulary

PIPELINE FRAMEWORKS:

PyTorch

HuggingFace Transformer

Transfer Learning Strategies

Full Fine Tuning

- All model parameters are updated
- High computational cost
- Risk of overfitting and losing pretrained knowledge

Full Fine-Tuning

Updates all 163 million parameters

Transfer Learning Strategies

LoRA

Updates only 0.8 million parameters

Paramter-Efficient Fine Tuning(PEFT)

using LoRA

- Only a small subset of parameters is trained (~1%)
- Faster and more memory efficient
- Preserves pretrained knowledge
- LoRA reduces training cost while maintaining strong performance

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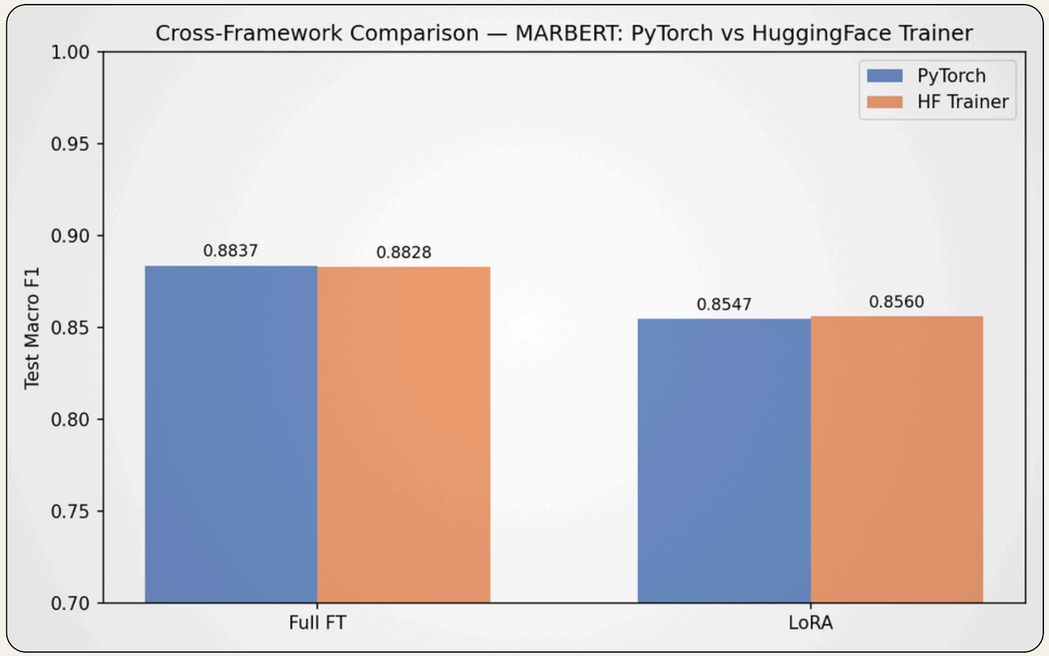
Full Fine-Tuning:    W' = W + ΔW        (163M params updated)

LoRA:                W' = W + (α/r) · B·A
                     where A ∈ R^(r×k),  B ∈ R^(d×r)
                     r = 8,  α = 16,  trainable: ~0.8M params (0.49%)
```

Experiments and Results



model	pipeline	test_macro_f1	test_accuracy	trainable_params	best_val_f1
MARBERT FFT	PyTorch	0.8837	0.9005	163000000	0.8923
MARBERT LoRA	PyTorch	0.8547	0.8719	297219	0.8653

model	strategy	framework	test_macro_f1	test_accuracy
MARBERT	Full FT	HF Trainer	0.8828	0.8998
MARBERT	LoRA	HF Trainer	0.856	0.8717

Trainable Parameters: 162M TO ~300K (548x reduction)

CSS Cultural Metric

- Traditional metrics (Accuracy, F1) only measure correctness
- Do not capture true understanding of mixed-language text
- Models may rely more on English than Arabic

Method:

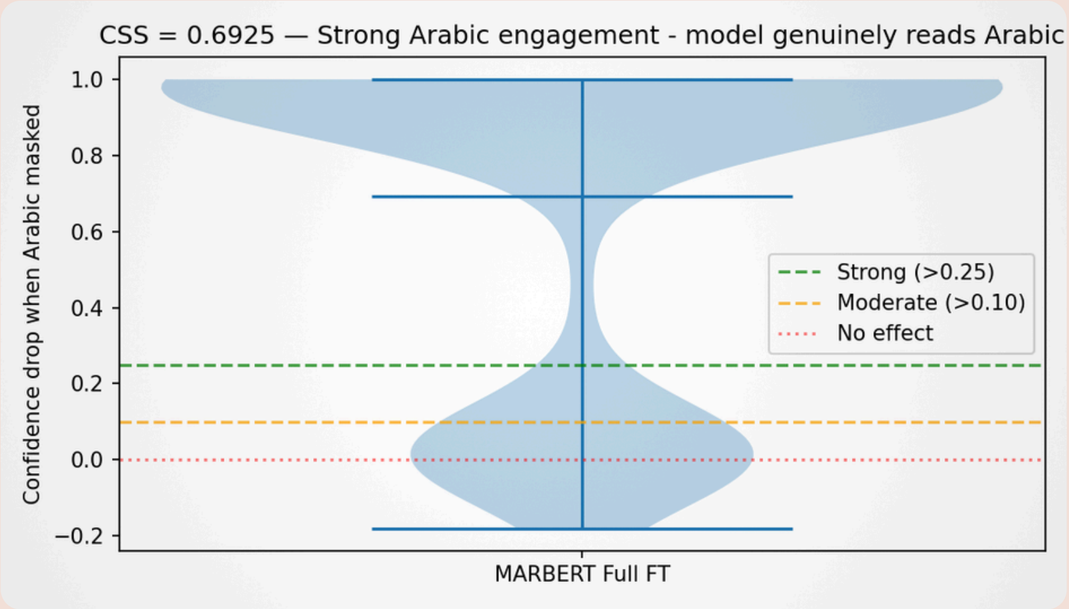
- Select correct predictions with high confidence (>80%)
- Remove Arabic text from input
- Re-run model and compare confidence

Interpretation:

CSS > 0.25 → Strong Arabic engagement

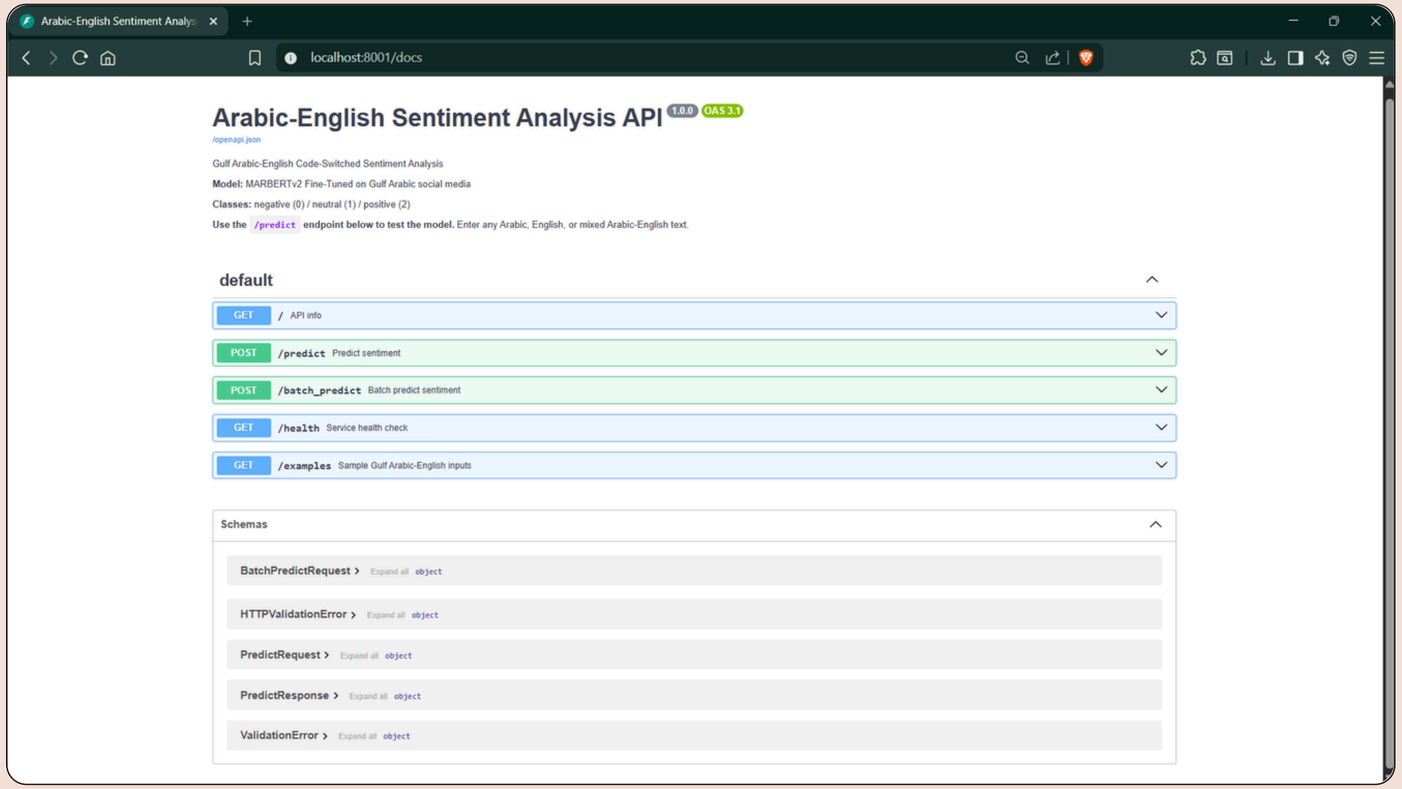
CSS 0.10–0.25 → Moderate engagement

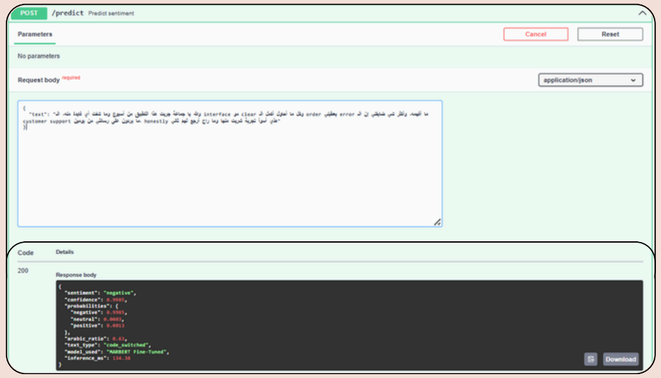
CSS < 0.10 → Weak (English-dependent)

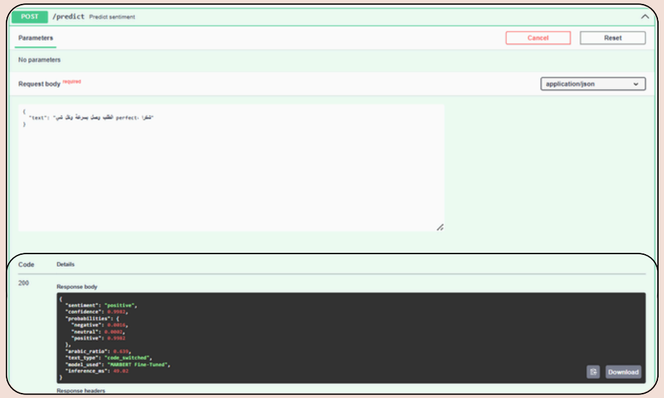



```
{
  "model": "MARBERT Full Fine-Tuning",
  "css_score": 0.6925,
  "interpretation": "Strong Arabic engagement - model genuinely reads Arabic",
  "n_samples_used": 355,
  "n_skipped_no_arabic": 0,
  "n_skipped_wrong": 126,
  "n_skipped_low_conf": 19
}
```

Deployment


Arabic-English Sentiment Analys
localhost:8001/docs
Arabic-English Sentiment Analysis API
1.0.0
OAS 3.1
/openapi.json
Gulf Arabic-English Code-Switched Sentiment Analysis
Model: MARBERTv2 Fine-Tuned on Gulf Arabic social media
Classes: negative (0) / neutral (1) / positive (2)
Use the /predict endpoint below to test the model. Enter any Arabic, English, or mixed Arabic-English text.
default
GET / API info
POST /predict Predict sentiment
POST /batch_predict Batch predict sentiment
GET /health Service health check
GET /examples Sample Gulf Arabic-English inputs
Schemas
BatchPredictRequest
HTTPValidationError
PredictRequest
PredictResponse
ValidationError


POST /predict Predict sentiment
Parameters
Cancel
Reset
No parameters
Request body
application/json
Code
Details
200
Response body
Download


POST /predict Predict sentiment
Parameters
Cancel
Reset
No parameters
Request body
application/json
Code
Details
200
Response body
Download
Response headers

Failure Analysis

What Worked:

- Transfer learning captured sentiment in Arabic and English
- Common sentiment words correctly identified
- Basic sentence structures handled well

Failure Cases:

- Gulf dialect vocabulary gaps (e.g., informal words)
- Difficulty detecting sarcasm in code-switched text
- Neutral class harder to classify

Negation

- Input: "this app is not at all bad"
- Expected: positive
- Predicted: negative (99.3% confidence)
- Why: model detects "bad" as strong negative, insufficient negation handling
- This is a field-wide limitation - not specific to our model

Data Limitations:

- Heavy use of synthetic data
- Less realistic than real-world text
- Reduced generalization ability

Conclusion

- Transfer learning proved effective for Gulf Arabic–English code-switched NLP despite limited labeled data.
- MARBERTv2 was adapted using both full fine-tuning and LoRA.
- Models pretrained on Arabic-rich data captured regional and dialectal patterns better than broad multilingual models.
- The Code-Switch Sensitivity Score (CSS) helped measure whether models truly understood Arabic in mixed-language text rather than relying mainly on English cues.
- Overall, the study shows that transfer learning is a practical solution for low-resource mixed-language tasks, though dialect and cultural nuance remain challenging.